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TUGAS MATEMATIKA PERTEMUAN 2

① a. $3,12131213\dots$

$$\Rightarrow a = 3,12131213\dots$$

$$a \cdot 10000 = 31213,12131213\dots$$

$$10.000a - a = 31213,12131213\dots - 3,12131213\dots$$

$$9999a = 31210$$

$$a = \frac{31210}{9999} \Rightarrow \text{Terbukti rasional}$$

b. $10,199999\dots$

$$\Rightarrow b = 10,199999\dots$$

$$b \cdot 100 = 1019,9999\dots$$

$$b \cdot 10 = 101,9999\dots$$

$$100b - 10b = 1019,9999\dots - 101,9999\dots$$

$$90b = 918$$

$$b = \frac{918}{90} = \frac{51}{5} \Rightarrow \text{Terbukti rasional}$$

② a. $-7x - 3 < 2x - 6$

$$-7x - 3 - 2x < 2x - 6 - 2x$$

$$-9x - 3 < -6$$

$$-9x - 3 + 3 < -6 + 3$$

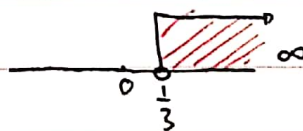
$$-9x < -3$$

$$\frac{-9x}{-9} > \frac{-3}{-9}$$

$$x > \frac{1}{3}$$

$$x > \frac{1}{3}$$

$\Rightarrow$



$$HP = \left(\frac{1}{3}, \infty\right)$$

$$HP = \left\{x \mid x > \frac{1}{3}, x \in \mathbb{R}\right\}$$

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b. $-x^3 + 4 \geq 3x^2$

$-x^3 - 3x^2 + 4 \geq 3x^2 - 3x^2$

$-x^3 - 3x^2 + 4 \geq 0$

$\Rightarrow -2 \mid \begin{array}{cccc} -1 & -3 & 0 & 4 \\ & 2 & 2 & -4 \\ \hline -1 & -1 & 2 & 0 \end{array} +$

$(x+2)(2x^2+2x-4)$

$(x+2)(2x+2)(x-1)$

$x+2=0 \mid 2x+2=0 \mid x-1=0$
 $x=-2 \mid 2x=-2 \mid x=1$
 $x=-\frac{2}{2}$
 $x=-1$

$\Rightarrow * x=2 \rightarrow -(2)^3 - 3(2)^2 + 4 \geq 0$

$-8 - 12 + 4 \geq 0$

$-16 \geq 0$ tidak memenuhi

$* x=-2 \rightarrow -(-2)^3 - 3(-2)^2 + 4 \geq 0$

$8 - 12 + 4 \geq 0$

$0 \geq 0$ memenuhi

$* x=-1 \rightarrow -(-1)^3 - 3(-1)^2 + 4 \geq 0$

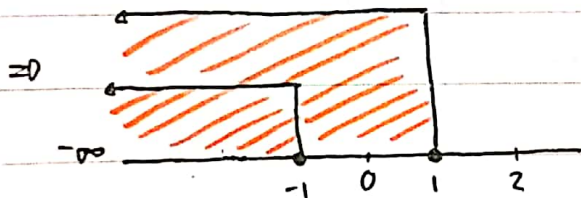
$1 - 3 + 4 \geq 0$

$2 \geq 0$ memenuhi

$* x=1 \rightarrow -(1)^3 - 3(1)^2 + 4 \geq 0$

$-1 - 3 + 4 \geq 0$

$0 \geq 0$ memenuhi



HP: $(-\infty, 1]$

HP: $\{x \mid -\infty < x \leq 1, x \in \mathbb{R}\}$

3) a. $\frac{2x}{x-1} > 3$

$$\frac{2x}{x-1} - 3 > 3 - 3$$

$$\frac{2x}{x-1} - 3 > 0$$

$$\frac{2x}{x-1} - \frac{3(x-1)}{x-1} > 0$$

$$\frac{2x - (3x-3)}{x-1} > 0$$

$$\frac{2x - 3x + 3}{x-1} > 0$$

$$\frac{-x+3}{x-1} > 0 \quad \begin{matrix} x=3 \\ x=1 \end{matrix}$$

$$\Rightarrow * x=2 \rightarrow \frac{-2+3}{2-1} > 0$$

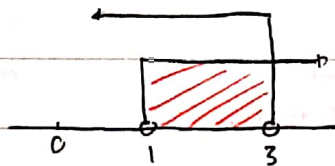
$$\frac{1}{1} > 0$$

$1 > 0$ memenuhi

$$* x=4 \rightarrow \frac{-4+3}{4-1} > 0$$

$$-\frac{1}{3} > 0 \text{ tidak memenuhi}$$

$\Rightarrow$



$$\text{HP: } (1, 3)$$

$$\text{HP: } \{x \mid 1 < x < 3, x \in \mathbb{R}\}$$

b. $|5 - \frac{2}{x}| \geq 1 \Rightarrow x = 5 - \frac{2}{x} \cdot a = 1$

$$|x| \geq a \Leftrightarrow x \leq -a \text{ atau } x \geq a$$

$$\Rightarrow x \leq -a$$

$$5 - \frac{2}{x} \leq -1$$

$$5 - \frac{2}{x} + 1 \leq -1 + 1$$

$$6 - \frac{2}{x} \leq 0$$

$$\frac{6x}{x} - \frac{2}{x} \leq 0$$

$$\frac{6x-2}{x} \leq 0 \Rightarrow \begin{matrix} 6x-2=0 \\ 6x=2 \\ x=\frac{2}{6}=\frac{1}{3} \end{matrix} \quad \begin{matrix} x=0 \rightarrow x \neq 0 \end{matrix}$$

$$* x = -1 \rightarrow \frac{6(-1)-2}{-1} \leq 0$$

$$\frac{-6-2}{-1} \leq 0$$

$$\frac{-8}{-1} \leq 0$$

$$8 \leq 0 \text{ tidak memenuhi}$$

$$* x = \frac{1}{3} \rightarrow \frac{6(\frac{1}{3})-2}{\frac{1}{3}} \leq 0$$

$$\frac{2-2}{\frac{1}{3}} \leq 0$$

$$\frac{0}{\frac{1}{3}} \leq 0$$

$$0 \leq 0 \text{ memenuhi}$$

$$* x = \frac{1}{6} \rightarrow \frac{6(\frac{1}{6})-2}{\frac{1}{6}} \leq 0$$

$$\frac{1-2}{\frac{1}{6}} \leq 0$$

$$-1 \cdot \frac{6}{1} \leq 0$$

$$-6 \leq 0 \text{ memenuhi}$$



$$\text{HP: } [0, \frac{1}{3}]$$

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$$\Rightarrow x \geq 0$$

$$5 - \frac{2}{x} \geq 1$$

$$5 - \frac{2}{x} - 1 \geq 1 - 1$$

$$4 - \frac{2}{x} \geq 0$$

$$\frac{4x-2}{x} \geq 0$$

$$\frac{4x-2}{x} \geq 0 \Rightarrow \begin{matrix} 4x-2=0 \\ 4x=2 \\ x=\frac{2}{4}=\frac{1}{2} \end{matrix} \quad \bigvee \quad x=0 \rightarrow x \neq 0$$

$$* x = -1 \rightarrow \frac{4(-1)-2}{-1} \geq 0$$

$$\frac{-4-2}{-1} \geq 0$$

$$\frac{-6}{-1} \geq 0$$

$$6 \geq 0 \text{ memenuhi}$$

$$* x = \frac{1}{4} \rightarrow \frac{4(\frac{1}{4})-2}{\frac{1}{4}} \geq 0$$

$$\frac{1-2}{\frac{1}{4}} \geq 0$$

$$\frac{-1}{\frac{1}{4}} \geq 0$$

$$-1 \cdot \frac{4}{1} \geq 0$$

$$-4 \geq 0 \text{ tidak memenuhi}$$

$$* x = \frac{1}{2} \rightarrow \frac{4(\frac{1}{2})-2}{\frac{1}{2}} \geq 0$$

$$\frac{2-2}{\frac{1}{2}} \geq 0$$

$$\frac{0}{\frac{1}{2}} \geq 0$$

$$0 \geq 0$$

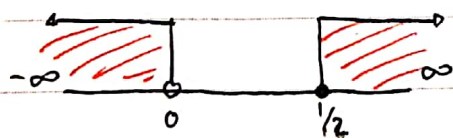
$$\text{memenuhi}$$

$$* x = 1 \rightarrow \frac{4(1)-2}{1} \geq 0$$

$$\frac{4-2}{1} \geq 0$$

$$\frac{2}{1} \geq 0$$

$$2 \geq 0 \text{ memenuhi}$$



$$\text{HIP} = (-\infty, 0) \cup [\frac{1}{2}, \infty)$$

$$\Rightarrow \text{HIP} = (-\infty, 0) \cup (0, \frac{1}{2}] \cup [\frac{1}{2}, \infty)$$

$$c. |4x+5| \leq 10$$

$$x = 4x+5, a = 10$$

$$|x| \leq a \Leftrightarrow -a \leq x \leq a$$

$$-10 \leq 4x+5 \leq 10$$

$$\Rightarrow -10 \leq 4x+5$$

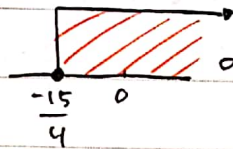
$$-10 - 5 \leq 4x+5 - 5$$

$$-15 \leq 4x$$

$$-\frac{15}{4} \leq \frac{4x}{4}$$

$$-\frac{15}{4} \leq x$$

$$x \geq -\frac{15}{4}$$



$$HP = [-\frac{15}{4}, \infty)$$

$$HP = \{x | x \geq -\frac{15}{4}, x \in \mathbb{R}\}$$

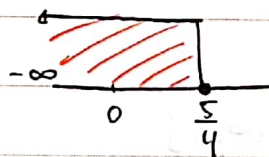
$$\Rightarrow 4x+5 \leq 10$$

$$4x+5 - 5 \leq 10 - 5$$

$$4x \leq 5$$

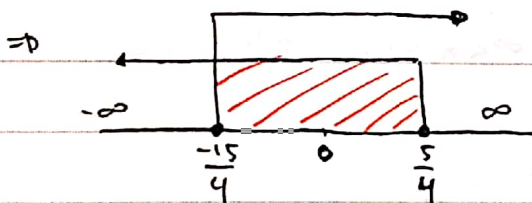
$$\frac{4x}{4} \leq \frac{5}{4}$$

$$x \leq \frac{5}{4}$$



$$HP = (-\infty, \frac{5}{4}]$$

$$HP = \{x | x \leq \frac{5}{4}, x \in \mathbb{R}\}$$



$$HP = [-\frac{15}{4}, \infty) \cup (-\infty, \frac{5}{4}]$$

$$HP = \{x | -\frac{15}{4} \leq x \leq \frac{5}{4}, x \in \mathbb{R}\}$$